

19. $\begin{bmatrix} 3 \\ -8 \\ 1 \end{bmatrix}, \begin{bmatrix} 6 \\ 2 \\ -5 \end{bmatrix}$

Solution: Let, $A = \begin{bmatrix} 3 & 6 \\ -8 & 2 \\ 1 & -5 \end{bmatrix}$. To check whether the columns of A is a basis of $\mathbb{R}^3$, we need to verify whether the columns of A are linearly independent and they also span $\mathbb{R}^3$.

$$\begin{aligned} \begin{bmatrix} 3 & 6 \\ -8 & 2 \\ 1 & -5 \end{bmatrix} &\xrightarrow[\times 1/3]{\text{Row1}=\text{Row1}} \begin{bmatrix} 1 & 2 \\ -8 & 2 \\ 1 & -5 \end{bmatrix} \xrightarrow[\text{Row2}=\text{Row2}]{+8 \times \text{Row1}} \begin{bmatrix} 1 & 2 \\ 0 & 18 \\ 1 & -5 \end{bmatrix} \xrightarrow[\text{Row3}=\text{Row3}]{-\text{Row1}} \begin{bmatrix} 1 & 2 \\ 0 & 18 \\ 0 & -7 \end{bmatrix} \\ &\xrightarrow[\times 1/18]{\text{Row2}=\text{Row2}} \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 0 & -7 \end{bmatrix} \xrightarrow[\text{Row3}=\text{Row3}]{+7 \times \text{Row2}} \begin{bmatrix} 1 & 2 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \xrightarrow[\text{Row1}=\text{Row1}]{-2 \times \text{Row2}} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \end{aligned}$$

It's very easy to verify that the final matrix is in rref. ~~There~~ There are no pivots in Column 3. A has only 2 pivots (Row1 Col 1, Row2 Col 2).

By the Invertible Matrix Theorem and the equivalence of the statements (c), (e) and (h), we know that the columns of A do not form a linearly independent set and they do not span $\mathbb{R}^3$.

Wrong logic.

Q2
By Theorem 4 from Chapter 1 and the equivalence of the statements (d) and (c), the columns of A do not span $\mathbb{R}^3$.
 $\therefore$ The given vectors cannot be a basis of $\mathbb{R}^3$ $\square$

20. $\begin{bmatrix} 1 \\ -6 \\ -7 \end{bmatrix}, \begin{bmatrix} 3 \\ -4 \\ 7 \end{bmatrix}, \begin{bmatrix} -2 \\ 7 \\ 5 \end{bmatrix}, \begin{bmatrix} 0 \\ 8 \\ 9 \end{bmatrix}$

Solution: Let, $A = \begin{bmatrix} 1 & 3 & -2 & 0 \\ -6 & -4 & 7 & 8 \\ -7 & 7 & 5 & 9 \end{bmatrix}$. To check whether the columns of A is a basis of $\mathbb{R}^3$, we need to verify whether

the columns of A are linearly independent and they also span $\mathbb{R}^3$.

The columns of A are linearly independent if and only if the matrix equation $Ax=0$ has only the trivial solution.